LLMs Know My Country Better Now, But Not Yet Good Enough

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Abstract

Large language models (LLMs) increasingly serve as critical information infrastructures for global development and policy decisions. However, whether rapid advances in LLMs have reduced or reinforced geographic disparities in factual recall remains unclear. We conduct a longitudinal assessment by updating WORLDBENCH through 2025 across 11 development indicators and 216 countries and territories, systematically evaluating an extensive suite of 33 earlier and recent LLMs across over 140K queries. We find that while recent LLMs reduce global mean error by 41.0%, geographic disparities persist, with errors remaining 1.58× as high in Sub-Saharan Africa as in North America and 1.27× as high in low-income as in high-income nations. Furthermore, open-source LLMs achieve substantial improvements yet their regional disparities remain largely unchanged, while proprietary LLMs exhibit lower errors and smaller geographic gaps. Such improvements are also uneven across development indicators, with marked gains in population and GDP but persistent stalls in critical welfare indicators such as maternal mortality and education expenditure. Enabling external retrieval serves as an effective path toward narrowing these gaps. Our findings reveal a persistent disconnect between advances in LLM factual accuracy and progress toward more geographically equitable access to socioeconomic knowledge.

1 Introduction

Large Language Models (LLMs) are rapidly becoming part of global knowledge infrastructures, increasingly relied upon to inform public policy, guide educational initiatives, and shape how individuals and institutions comprehend socioeconomic realities (Tsvetkova et al., 2024; UNDP, 2025). However, a growing body of evidence shows that LLMs exhibit severe geographic disparities when recalling basic socioeconomic facts,

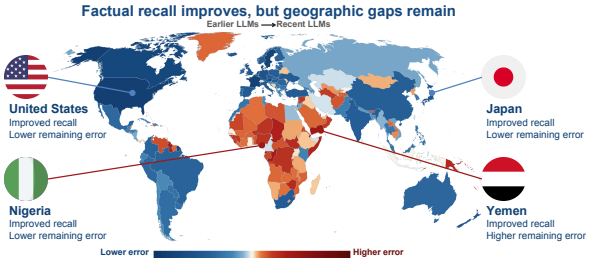


Figure 1: **Geographic disparities in LLM factual recall.** Errors are highest across developing nations (e.g., Nigeria and Yemen) and lowest in advanced economies (e.g., the US and Japan).

performing particularly poorly on Least Developed Countries (LDCs) and low-income economies (Manvi et al., 2024; Decoupes et al., 2025). For example, the benchmark WORLDBENCH found that early LLMs suffered error rates up to 1.5× higher on Sub-Saharan Africa and lower-income nations than on Western regions (Moayeri et al., 2024). Such disparities extend beyond simple fact-checking. When propagated into downstream reasoning and decision-making, these inaccuracies may pose acute risks of misguiding grassroots policy analyses, misallocating developmental assistance, and further reinforcing existing inequity in social development (World Bank, 2025).

Recent substantial efforts have introduced various techniques to improve LLMs' capability of factual recall, including knowledge alignment, parametric editing, and retrieval augmentation (Lin et al., 2024; Fang et al., 2025; Vu et al., 2024). Iterative updates across both proprietary and open-source LLMs have advanced on general capability benchmarks (Yang et al., 2025). However, it remains an open and urgent question whether these rapid advances have truly mitigated geographic disparities in factual recall. As illustrated in Figure 1, if scaling laws and architectural improvements primarily benefit data-rich countries, rising average accuracy may conceal persistent, or even widen-

ing, geographic inequities in access to socioeconomic facts between central and peripheral regions (Godey et al., 2024; Coelho et al., 2025). To investigate whether modern AI progress bridges this uneven development, we attempt to ask **How do recent state-of-the-art LLMs perform overall on global factual recall compared to earlier LLMs?** and **Have performance gaps across regions and income groups narrowed proportionally, or has relative unfairness persisted?**

In this paper, we aim to examine how geographic disparities in LLM factual recall have evolved across model generations over the past two years. First, we update the WORLDBENCH benchmark with the latest World Bank and UN development statistics through 2025 across all eleven indicators, establishing an updated ground truth covering 216 countries and territories. Second, we curate an extensive evaluation suite of over 30 earlier and recent LLMs, contrasting open-source with leading proprietary LLMs, and tracking generational changes within the Llama and Qwen families at a controlled parameter scale. Third, our empirical analysis reveals several key insights:

- **Geographic disparities persist despite general progress.** Despite a 41.0% global error reduction, modern LLMs still incur 1.58× as high in Sub-Saharan Africa as in North America and 1.27× as high in low-income as in high-income nations.
- **Open-source LLMs' gains fail to narrow regional disparities.** Open-source LLMs steadily lower errors yet their regional disparity remains largely stagnant, while proprietary LLMs have both lower error and smaller geographic gaps.
- **Progress is uneven across development indicators.** Accuracy concentrates in macroeconomic aggregates like population and GDP, while life-critical welfare indicators like maternal mortality and education expenditure improve more slowly.
- **Web search bridges the capability divide.** Enabling web search allows lightweight open-source LLMs to rival the factual accuracy of leading proprietary LLMs.

1.1 Geographic and Socioeconomic Disparities in LLMs

Algorithmic fairness in LLMs is increasingly intertwined with global social development (Mohamed et al., 2020). Extensive studies have demonstrated that LLMs' intelligence and their benefits remain uneven across geographic regions and socioeconomic groups (Moayeri et al., 2024; Manvi et al., 2024). For instance, LLMs often exhibit socioeconomic biases in spatial reasoning (Tanwar et al., 2025a), travel advice contains less location information for poorer nations (Bhagat et al., 2025), and narratives set in these countries tend to emphasize hardship and poverty (Bhagat et al., 2025). Similar imbalances emerge in regional fact-checking (Coelho et al., 2025) and cross-lingual access to culturally specific factual knowledge (Tanwar et al., 2025b). In particular, WORLDBENCH quantified this gap in factual recall using standardized World Bank statistics, revealing that early LLMs exhibited error rates approximately 1.5 times higher for Sub-Saharan Africa and low-income countries than for Western regions (Moayeri et al., 2024). These geographical disparities may propagate directly into downstream complex reasoning, risking the distortion of development priorities, resource allocation, and policy decisions (World Bank, 2021). Thus, our study aims to answer whether rapid LLMs' advancement naturally narrows these disparities and improves fairness in factual recall.

1.2 Factual Knowledge and Temporal Reliability in LLMs

Factual recall is a crucial capability for LLMs to function as reliable information infrastructure, underpinning various downstream applications ranging from education to policy reasoning (Wei et al., 2024; Fatahi Bayat et al., 2025). However, the parametric memory of LLMs is limited by imbalances in training data, and they are prone to hallucinations when handling long-tail entities or evolving socioeconomic facts (Kandpal et al., 2023; Vu et al., 2024). Since real-world statistical facts shift over time, static evaluations inevitably suffer from temporal misalignment and benchmark lag, thereby distorting performance comparisons across model generations (Zhu et al., 2025; Jiang et al., 2026).

To mitigate these, recent research has sought to enhance factual accuracy through techniques such as factuality-aware alignment (Lin et al., 2024; Huang and Chen, 2024), parametric knowledge editing (Fang et al., 2025), and retrieval-augmented generation (RAG) (Vu et al., 2024). Yet, these technical breakthroughs have also raised a critical fairness challenge. Since overall improvements in factual accuracy are primarily driven by optimization against and evaluation on benchmarks

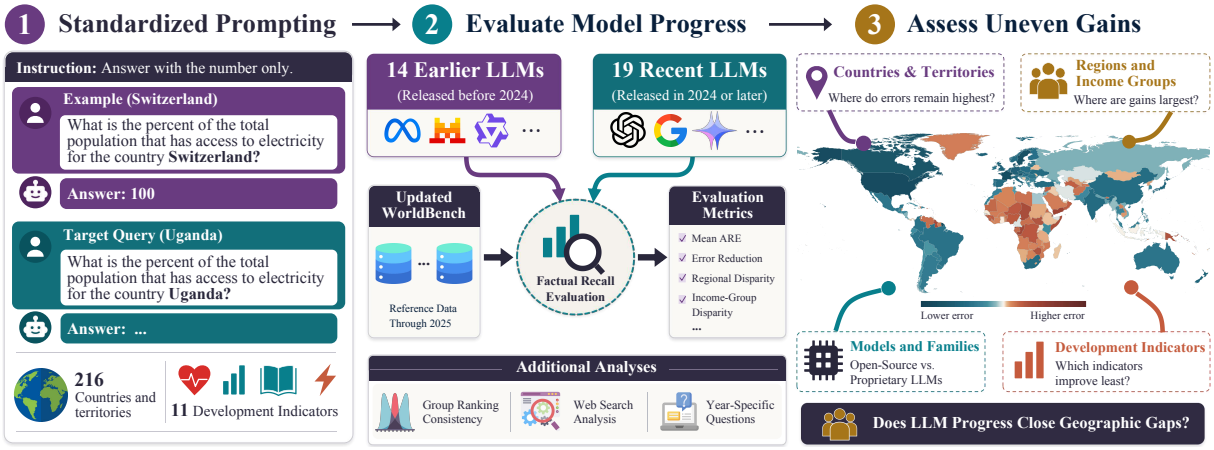


Figure 2: **Overview of the study framework.** We deploy standardized one-shot prompting across 216 countries and 11 development indicators (left), benchmark 33 earlier and recent LLMs against updated reference statistics through 2025 (middle), and audit factual disparities across geographic regions, income groups, model&families, and development indicators (right).

(Fatahi Bayat et al., 2025), they may disproportionately reinforce well-documented entities while yielding negligible gains to underrepresented contexts (Godey et al., 2024). Consequently, such advancements risk concealing persistent, or even widening, disparities faced by data-scarce regions or developing nations. While prior literature has examined geographic disparities (Decoupes et al., 2025; Manvi et al., 2024), there is a lack of in-depth research on how these disparities evolve across model generations. This study aims to assess the performance of recent state-of-the-art LLMs on national development facts and to investigate whether the rapid scaling of algorithms over the past two years has democratized factual reliability or perpetuated historical inequalities.

2 Methods

We conduct a longitudinal assessment by updating WorldBench (Moayeri et al., 2024) through 2025 across 11 development indicators and 216 countries and territories, systematically evaluating an extensive suite of 33 earlier and recent LLMs in factual accuracy and geographic disparities. Figure 2 outlines our study framework.

2.1 Dataset and Country Categories

Updated WorldBench We construct an updated benchmark grounded in WorldBench's eleven core socioeconomic indicators. The indicators span diverse socioeconomic development areas, covering population, economic activity, health, public services, political representation, land use, and energy (see Appendix A.1 for full definitions and units). To assess the models' knowledge of the latest contemporary facts, we extend the reference time window to 2023–2025, drawing primarily from the World Bank's World Development Indicators (WDI), supplemented by renewable energy data from the UN Statistics Division's Sustainable Development Goals (SDG) database (World Bank, 2026; United Nations Statistics Division, 2026). Regarding data selection and processing, we exclude regional aggregated data and retain only those countries with at least one valid observation record within this window. Furthermore, to address the issue of unbalanced available data in the annual data reports, we use the mean of all available annual observations over the past three years (2023–2025) as the ground truth. The final evaluation benchmark comprises a total of 2,112 valid "country-indicator" queries covering 216 countries and regions. The updated data are available at `https://github.com/nicezheng/WorldBench_2023_2025_Benchmark`.

Country Categorization To examine whether recent models narrow performance gaps across geographic regions and income levels, we adopt the World Bank's country classifications. This taxonomy partitions the world into 7 regions (East Asia & Pacific, Europe & Central Asia, Latin America & Caribbean, Middle East & North Africa, North America, South Asia, and Sub-Saharan Africa) and 4 income groups (High, Upper-Middle, Lower-Middle, and Low Income), mapped in Appendix A.2. All cross-group comparisons use these

fixed categories, providing an objective, externally defined standard to assess performance disparities.

2.2 Model Selection

We curate an expansive evaluation suite of 33 representative LLMs grouped by release period: 14 earlier models (2023 or before) and 19 recent models (2024 onward). Appendix B lists the details for providers, versions, and inference pipelines. The earlier set retains available open-source LLMs from 6 families examined in WORLDBENCH: Llama 2, Qwen, Orca 2, Vicuna, Mistral, and Zephyr. Early proprietary LLMs are excluded because most providers have deprecated or ceased maintaining those earlier versions. The recent set includes 12 open-source and 7 proprietary LLMs, covering different model families, scales, and developer origins. Alongside major US and Chinese developers, we intentionally incorporate models from diverse regional ecosystems, including Canada's Cohere, South Korea's Upstage, and the UAE-led Falcon and Jais projects. We also analyze open-source and proprietary LLMs separately, and conduct within-family comparisons across successive generations of Qwen and Llama with model scale controlled at approximately 7B–9B parameters. Local open-source models are executed via vLLM, while larger models and proprietary LLMs are queried through official OpenRouter APIs ().

2.3 Experimental Design

Task Formulation We formulate country-level factual recall as a closed-book querying task where an LLM is asked to report a specific development indicator for a target nation. Because annual reporting by international agencies is often lagged or varies across countries, directly matching a single year introduces missing-data bias. Following WORLDBENCH, we address this by taking the average of all available annual observations from 2023 to 2025 as the ground truth for each country and indicator. To complement this contemporary evaluation and directly compare earlier and recent models against an identical annual baseline, we also conduct a supplementary year-specific experiment querying facts explicitly for the year 2021 (detailed in Appendix D). Across 33 evaluated LLMs and 2,071 eligible country–indicator pairs, the primary evaluation generates 68,343 total queries, totaling over 125,000 queries across all experiments.

Prompting Design We implement a standardized, one-shot prompt to query each model (Appendix C.1). Each prompt comprises a base instruction requiring a concise numerical response, an example question and answer using Switzerland, and the target question specifying the country, development indicator, and statistical unit. Switzerland serves as a fixed formatting example and is strictly excluded from scoring. This same prompt format is applied across the main task, the year-specific experiment, and the web-search evaluation across 6 modern models.

Result Parsing Each model generates one designated format response per question, using official chat templates for instruction-tuned models and text completion for base models. To facilitate large-scale automated numerical extraction, we adopt a rule-based parser that handles thousands separators, scientific notation, explicit units, and magnitude terms (e.g., *million* and *billion*). The parser extracts candidate numbers purely from the text structure of the response without consulting ground truth to avoid selection bias. Responses that fail to provide a valid numerical value, such as refusals, API errors, and truncated outputs, are manually verified, labeled as unscorable, and recorded in answer coverage separately. All extraction rules and procedures are documented in Appendix C.2.

2.4 Error Metrics

We measure the discrepancy between a nonnegative model answer a and reference value y using absolute relative error (ARE):

$$\mathrm{ARE}(a, y) = \frac{|a - y|}{\max(a, y)}. \tag{1}$$

ARE ranges from zero to one, with lower values indicating higher accuracy. Dividing by $\max(a, y)$ normalizes magnitude differences, allowing fair comparison across different units and numerical scales. Mean ARE is calculated by pooling all valid responses within the corresponding groups.

To quantify uneven performance across groups, we also define inter-group disparity as the maximum gap between group means:

$$\text{Disparity} = \max_{g \in G} E_g - \min_{g \in G} E_g, \tag{2}$$

where E_g denotes the mean ARE for group g, calculated separately for the 7 regions and 4 income groups. Larger values indicate a wider gap

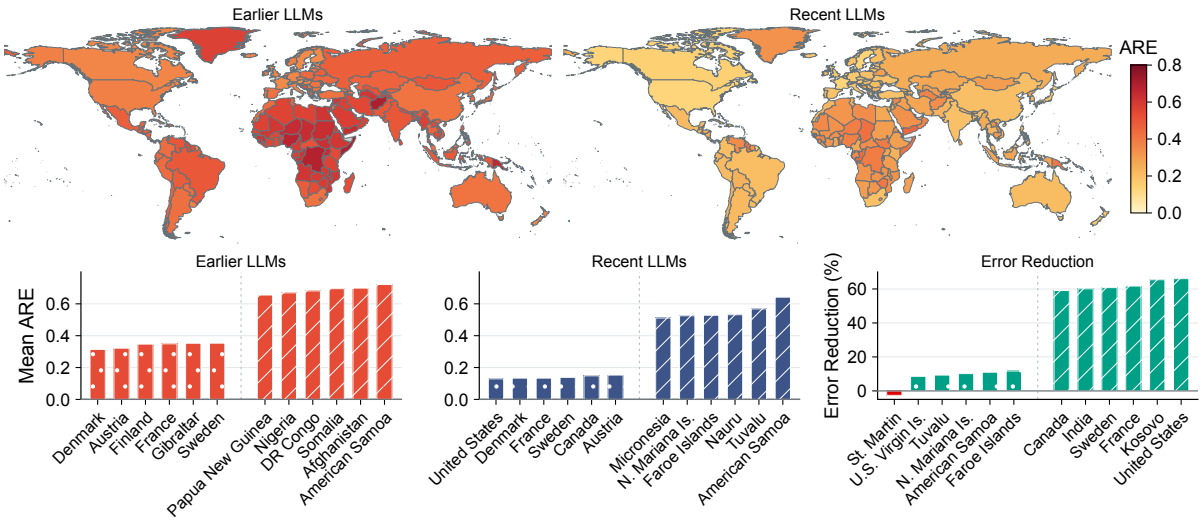


Figure 3: **Country mean errors and extremes for earlier and recent LLMs.** Top panel shows the error map for the earlier (left) and recent models (right). Lighter colors indicate lower error, and gray denotes no mapped score. Bottom panels show the six lowest and highest errors in each set and the smallest and largest percentage reduction.

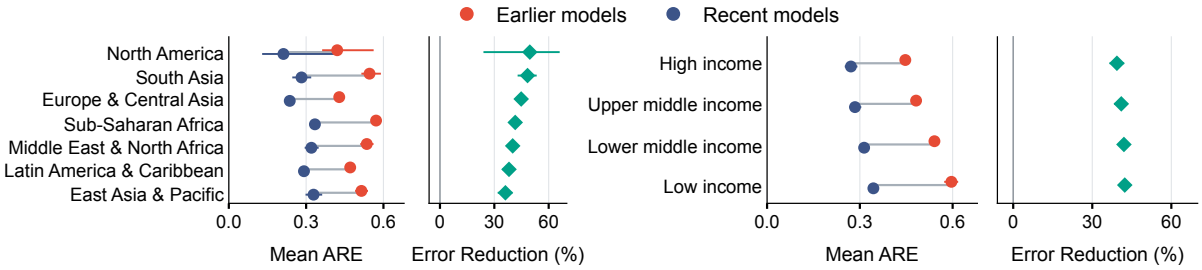


Figure 4: **Mean ARE and proportional error reduction by region and income group.** Groups are ordered by decreasing ARE reduction. Thin lines are 95% paired country bootstrap intervals.

between the best- and worst-performing groups. We quantify model progress in both mean error and disparity through percentage reduction, $100 \times (1 - E_{\text{recent}}/E_{\text{earlier}})\%$, where positive values indicate improvement. To assess statistical uncertainty, we calculate 95% confidence intervals using 5,000 paired country-level bootstrap samples. Finally, to determine whether observed disparities exceed chance, we report a random-grouping reference by averaging disparity across 10 random assignments of countries to matching group sizes.

3 Results

3.1 Overall geographic disparities

We find that across all evaluable country–indicator responses, mean ARE decreases from 0.5 for earlier models to 0.295 for recent models, a reduction of 41% (95% CI: 39.6% to 42.3%). As mapped in Figure 3, this improvement is nearly universal: 215 out of the 216 countries and regions exhibited a trend of improvement. However, even as error rates generally declined, the distribution of countries exhibiting extreme error shifted. While countries with the lowest errors consistently remain concentrated in North America and Western Europe (e.g., the United States, Denmark, France; mean ARE ≤ 0.15), the countries with the highest errors have shifted from low-income continental nations in earlier models (e.g., Afghanistan, the Democratic Republic of the Congo, Somalia; ARE > 0.65) to remote Pacific island economies in recent models (e.g., Tuvalu, American Samoa, Nauru; ARE > 0.50). Interestingly, the largest proportional gains are still concentrated in North America and Europe, led by the United States (66.1%) and Kosovo (65.4%). All countries with the smallest reductions are also island economies, e.g., Tuvalu's error falls by only 9.1%. Crucially, the ranking of countries by error rate remained highly correlated across model generations (Spearman $\rho = 0.772, p < 0.001$). These findings suggest that while the recent LLM progress has generally lowered fact recall errors across countries,

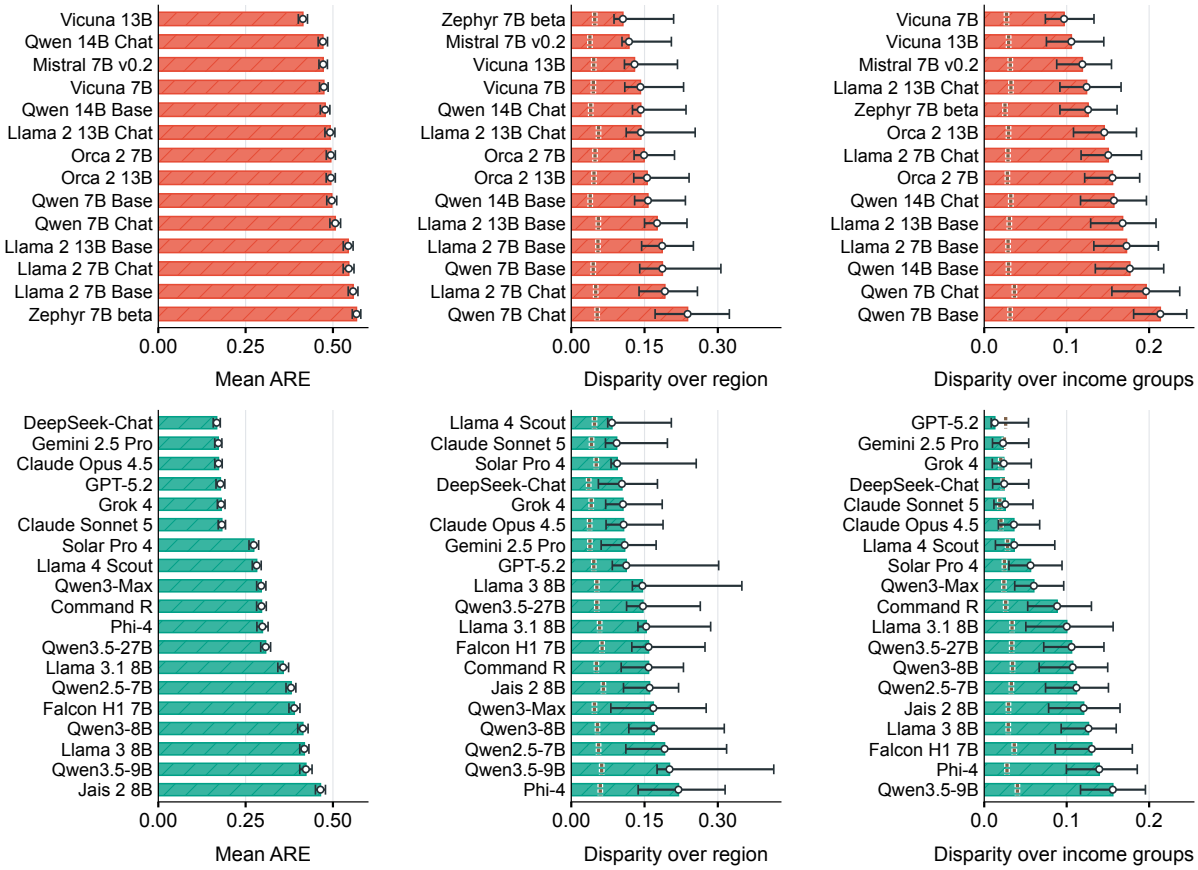


Figure 5: **Per-model mean ARE, disparity over regions, and disparity over income groups.** The top contains 14 earlier models and the bottom 19 recent models. Dark capped whiskers show the full, two-sided 95% country bootstrap intervals. Brown dashed markers show each model's disparity under random country categorization into 7 groups for regions and 4 for income groups, averaged over ten trials.

it preserves the geopolitical hierarchy founded in WORLDBENCH while retracting performance extremes into digitally sparse island economies.

3.2 Disparities across regions and income groups

Figure 4 demonstrates whether recent model developments have fostered performance convergence across different regions and income groups. Across regions, North America and Europe have consistently led in accuracy, while Sub-Saharan Africa has consistently recorded the lowest rates. The magnitude of improvement also varied highly across regions ranging from 49.6% in North America to 36.1% in East Asia and the Pacific. Interestingly, while the East Asia and Pacific region shows the slowest overall improvement, after checking specific countries, we find that major AI powerhouses like China and Japan actually saw gains of approximately 50%, and almost all of the marginal improvements were concentrated among small Pacific island nations. Across income groups, both earlier and recent LLMs exhibit a monotonic characteristic: factual error rates rise as national wealth levels decline. Surprisingly, the error reductions are almost identical across all four income groups ($\approx 42\%$), implying that the relative gap between rich and poor nations remained entirely unchanged.

3.3 Disparities across LLMs

Figure 5 illustrates per-model accuracy and disparities across model generations. The results show that a majority of recent LLMs exhibit marked improvements in mean ARE, with only a few smaller or specialized models, such as Jais-2-8B (mean ARE of 0.460), remaining at earlier error levels. Across all 33 evaluated LLMs, regional disparities universally remain above the random classification baseline, confirming that factual recall across geographic regions remains highly uneven across every model. In contrast, disparities across income groups have narrowed significantly among modern LLMs. Frontier models such as GPT-5.2 and Grok 4 have approached or even fallen below the random

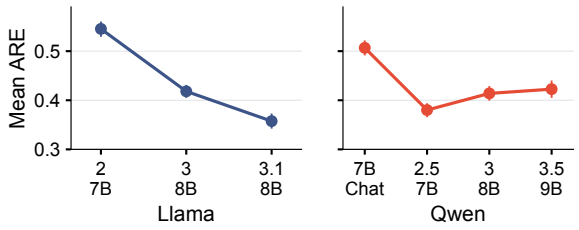


Figure 6: **Mean ARE for similarly sized Qwen- and Llama- family models.**

baseline, whereas most open-source models still exhibit substantial bias.

Open-source vs. Proprietary LLMs Proprietary LLMs hold a decisive advantage over open-source in both accuracy and fairness, achieving a lower mean ARE (0.207 vs. 0.348) alongside substantially smaller regional (0.108 vs. 0.151) and income-group disparities (0.027 vs. 0.102). In contrast, while open-source LLMs have reduced their mean error by 30.4% from 0.500 to 0.348, their regional disparity is almost unchanged across both sets, indicating that geographic inequities in open-source LLMs remain largely unaddressed.

Model family Comparison Figure 6 illustrates two distinct evolutionary paths taken by various model families within the 7B–9B parameter range. The Llama family demonstrates steady improvement across iterations, with mean ARE dropping from 0.545 for Llama 2 to 0.357 for Llama 3.1. In contrast, the Qwen family exhibited an unexpected trend of rising errors. While the error decreases from 0.503 to 0.375 in the Qwen2.5-7B version, it subsequently rises to 0.410 and 0.417 in the Qwen3-8B and Qwen3.5-9B versions, respectively. These results suggest that a higher model version does not necessarily guarantee greater reliability in accessing global factual socioeconomic information.

3.4 Disparities across indicators

Figure 7 shows that the 11 indicators are ordered by proportional error reduction. Recent models achieve large gains on macro-economic aggregates, with population and GDP errors dropping by 80.0% and 65.4%, respectively. However, progress in social welfare indicators critical to public well-being is more limited. Error reductions for maternal mortality, education expenditure, and the proportion of women in parliament are all under 30%. This suggests that LLMs have gradually mastered macro-economic and infrastructural facts, yet they lag far behind on essential human welfare. Besides, the development of recent LLMs may have also exacerbated disparities in factual recall. For instance, regional disparities in GDP per worker (PPP) have increased from 0.171 to 0.264, while the gaps in maternal mortality and GDP indicators across income groups have nearly doubled.

3.5 Consistency of Regional and Income-Group Orderings

The ranking heatmaps (see Appendix E.2) illustrate the consistency with which the models rank regions and income groups across various indicators. The results show that model accuracy in factual recall has solidified at the top while dispersing at the bottom. North America consistently achieves the lowest error across almost all models, reinforcing clear Western regions' dominance. In contrast, the lowest-ranked regions become more dispersed. While earlier LLMs performed worst mainly in Sub-Saharan Africa (11 of 14 models), recent LLMs distribute their highest error across East Asia and the Pacific (7 models), Sub-Saharan Africa (6 models), and the Middle East and North Africa (5 models). Across income groups, the disadvantage faced by poorer economies remains largely consistent: 16 out of the 19 recent models (and all 14 earlier models) show the highest error rates for low-income countries. Importantly, this socioeconomic disparity stems primarily from certain specific indicators rather than uniform failure across the board, as low-income countries ranked worst on only 6 of the 11 individual metrics. These findings indicate that modern LLMs maintain the highest accuracy in Western regions, while performance disparities are dispersed throughout the developing world and are dependent on specific evaluation metrics.

3.6 The impact of web search strategy

Figure 8 compares standard and web-search-enabled performance across six representative LLMs, demonstrating that external retrieval acts as a powerful tool in enhancing the recall of global factual information. Enabling web access significantly reduces mean ARE by 63.4% to 84.3%, narrowing the performance disparities across architectures and resulting in a tight final range of 0.058 to 0.067. Notably, lightweight open-source LLMs benefited the most. The ARE for Qwen3.5-9B fell from 0.417 to 0.065, almost eliminating the performance gap relative to proprietary LLMs like GPT-5.2 and Gemini 2.5 Pro. Simultaneously, web search reduced disparities across geographic regions and

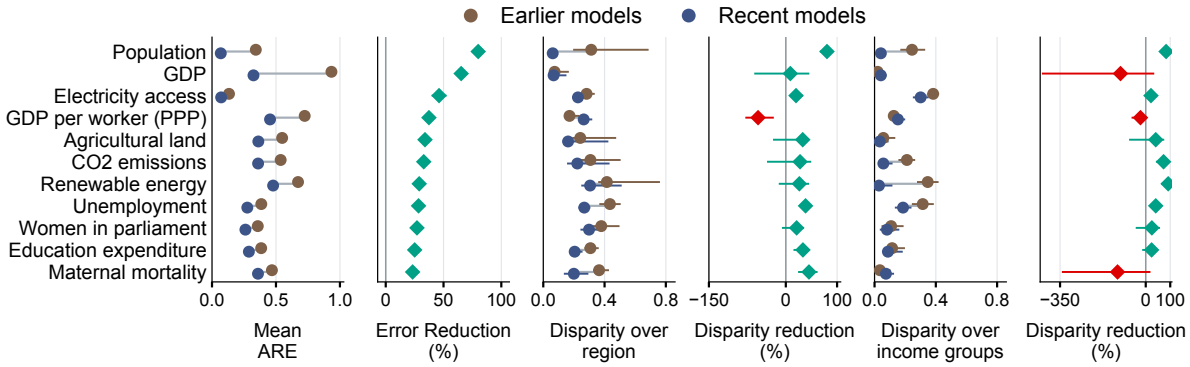


Figure 7: **Errors, geographic disparities and percentage reductions by indicator.** All panels follow decreasing ARE reduction. Negative disparity reductions indicate larger recent gaps. Thin lines are 95% paired country bootstrap intervals.

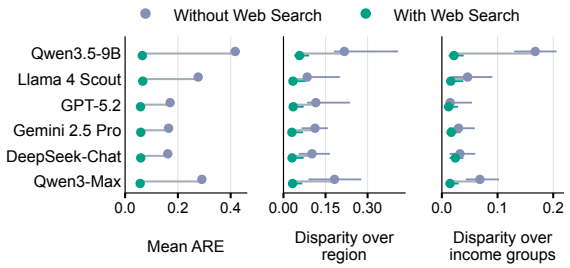


Figure 8: **Errors and disparities with and without Web Search.**

income groups for all evaluated LLMs, bringing geographic variance down to 0.03–0.06 and narrowing income-based disparities to near-zero levels. This demonstrates that equipping accessible models with web search is a practical pathway to narrow geographic knowledge gaps across model architectures without necessitating prohibitive parametric scaling.

4 Discussion

The societal value of LLMs largely depends on whether the benefits they generate can be equitably distributed across nations (Hosseini and Khanna, 2026). Our findings suggest that general improvement in accuracy often masks persistent geographic inequalities. Because models improve in near-proportionally across income groups, the relative knowledge gaps between wealthy and developing nations actually narrowed. This simultaneous improvement creates an illusion of progress, leaving disadvantaged nations in the same position of relative disadvantage in the AI era (Khan et al., 2024).

This persisted geographical disparities also reflects a deeper distortion in the geopolitical bias of artificial intelligence (Vergara, 2026; Li et al., 2024). Modern models excel at recalling macroeconomic aggregates like GDP, yet remain blind to life and welfare realities. When international organizations increasingly rely on LLMs for policy planning and humanitarian assistance, this bias risks prioritizing capital-driven agendas while obscuring urgent social crises in disadvantaged regions (World Bank, 2021). Besides, bridging the global divide in reliable access to knowledge cannot be achieved solely by scaling model parameters or refining architectures (Zhou et al., 2024). Open-source LLMs steadily lower error rates, yet their regional disparities remain stagnant, and they lag largely behind commercial proprietary LLMs in equity. Fortunately, our web-search findings show that grounding models in external evidence enables 9B parameter lightweight open-source LLMs to match the accuracy of frontier LLMs. For resource-constrained institutions or nations, building AI systems that enable web search is a more pragmatic path than competing in the costly race for raw computational power.

5 Conclusion

We conducted a longitudinal audit of geographic disparities in LLM factual recall across 33 models and 216 countries through 2025. Although recent LLMs reduce global error by 41.0%, relative geographic divides remain largely unchanged. Errors remain 1.58× as high in Sub-Saharan Africa as in North America, and 1.27× as high in low- versus high-income nations, with performance extremes retracting into data-scarce Pacific island nations. Moreover, gains favor macroeconomic aggregates over human welfare, and open-source LLMs lag in regional equity. While web search offers an effective equalizer across architectures, general model scaling alone fails to ensure equitable factual reliability for disadvantaged nations.

Limitations

This study has several limitations that define the scope of the empirical analysis and point toward future research directions. First, our evaluation relies on standardized English-language numerical queries, which may not fully capture colloquial expressions, multilingual contexts, or qualitative narratives. Nevertheless, constructing queries based on official international statistics provides an objective and verifiable "ground truth" covering 216 countries, thereby avoiding distortions caused by translation bias or subjective scoring. Second, querying statistical data without specifying a timestamp can introduce temporal ambiguity, as models might retrieve data from earlier reporting cycles. To balance the issue of inconsistent data availability across years, we adopted the valid average of the most recent three years as the ground truth. We also conducted supplementary experiments querying specific years across 27 models (see Appendix D); the results confirmed that disparities across regions and income groups persist regardless of whether temporal constraints are applied. Third, our findings reflect the state of the rapidly evolving model ecosystem as of 2026. Although model architectures and training corpora undergo continuous iteration, our comparative analysis of model families with similar parameter scales (such as the 7B–9B range for Llama and Qwen) reveals that these structural disparities remain consistent.

AI Declaration Statement

During the preparation of this manuscript, the authors utilized Gemini 3.1 Pro and GPT 6 Astra to support language polishing of the programmatic text blocks. Following the use of these tools, the authors thoroughly reviewed, critically evaluated, and edited the text to ensure absolute factual grounding and technical accuracy. The authors maintain full accountability and ownership over the final text, context, and structural analysis presented herein.

Acknowledgments

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Appendices

A Data and Country Classifications

A.1 Indicators and Reference Data

We compile the evaluation inventory separately for each indicator, excluding regional aggregates and reserving Switzerland for prompt examples. The resulting inventory contains 2,190 country–indicator pairs, of which 2,112 have at least one reference observation in 2023–2025. These eligible pairs span 216 countries and territories, with indicator-level coverage reported in Table 1. The other 78 pairs have no observations in this period, including 57 for education expenditure. For each eligible pair, we use the arithmetic mean of its available annual values in 2023–2025, without substituting observations from other years. The reference compilation incorporates the World Bank's 2026 data update and retains three-decimal precision for scoring.

A.2 Country Groupings

Figure 9 maps the seven regional and four income-group assignments inherited from WorldBench. We hold this classification fixed across all models and conditions. All 216 evaluated countries and territories have regional assignments, and 215 have income-group assignments. The income groups follow the WorldBench classification and are distinct from the United Nations' least-developed-country designation.

B Models Inventory and Experimental Coverage

Our main evaluation covers 33 models, comprising 14 earlier and 19 recent releases. Table 2 documents their developers, release records, model versions, and inference pipelines. The year-specific experiment uses a subset of 27 models (12 earlier and 15 recent), excluding several higher-cost proprietary models, including Claude Opus 4.5, Grok 4, Qwen3-Max, and Solar Pro 4. Table 3 reports mean ARE alongside the proportion of questions yielding scorable numerical responses. Of the 69,696 model–question combinations in the main task (33 models × 2,112 questions), 68,701 produce eligible answers. The year-specific task includes 56,404 eligible answers among 57,537 combinations (27 models × 2,131 questions).

C Prompting, Parsing, and Validation

C.1 Prompt Examples and Inference

Following WorldBench (Moayeri et al., 2024), we use one-shot prompting that specifies the indicator and its unit, provides a question–answer example for Switzerland, and requests the corresponding value for a target country. Switzerland is reserved exclusively for these examples. Table 4 reproduces the agricultural-land prompts used in the experiments. The main task retains the original Switzerland example. In the year-specific task, both the example and target question explicitly refer to 2021, and the example answer is drawn from Switzerland's 2021 observation.

C.2 Parsing and Scoring Checks

We extract numerical predictions with a parser that recognizes thousands separators, scientific notation, magnitude expressions, and units. It separates candidate answers from dates and unit denominators, and retains a prediction when a single nonnegative value can be identified. Explicit numerical estimates are treated in the same way as other answers. Refusals, empty responses, API failures, ambiguous numerical outputs, and truncated responses receive distinct extraction statuses and are excluded from numerical scoring.

The validation procedure checks model–question identifiers, country assignments, requested years, units, and extracted values. Records previously recovered with ground-truth assistance are excluded, after which errors are recomputed against the fixed reference values. Independent recalculations reproduce the reported mean errors, geographic disparities, response coverage, and confidence intervals.

C.3 Aggregation and Uncertainty

We calculate the average ARE by aggregating scorable model-question responses across subsets of reports, assigning equal weight to each response. Missing or unscorable answers affect model coverage but do not result in the exclusion of valid answers from other models. For individual models, geographic differences are calculated based on the mean for their region or income group; for model sets and metrics, differences are calculated based on aggregated group means rather than the average of model-specific differences. Country rankings utilize all scorable responses for each country, thereby reflecting the mix of metrics for which valid an-

swers are available. We report response coverage alongside error figures to illustrate the basis of the evaluation.

We employ 5,000 country-level bootstrap samples to estimate 95% percentile confidence intervals. Each sample retains all model and metric observations for a given country, and identical resampled sets are used for pairwise comparisons. When comparing regions and income groups, resampling is performed at the country level within the respective groups. For reference, we also randomly assign all countries—with equal probability—to seven regional or four income categories and average the resulting disparity values across ten trials; this randomization preserves the number of categories but does not maintain the observed group sizes.

D Year-Specifying Evaluation

To assess the ability to recall factual information specific to a particular year, we designed a dedicated task in which both the target questions and the Switzerland example focused on 2021. This task covered 216 countries and regions, involved 2,131 questions, and was evaluated using 12 earlier models and 15 recent models. We employed the same metric definitions, ARE (Average Relative Error) standards, and scoring aggregation methods used in the main task to score model responses against the corresponding 2021 observed values; all reference values were drawn exclusively from that specific year.

The evaluation results for the specific year show that the average ARE decreased from 0.438 to 0.299, a reduction of 31.8% (95% confidence interval: 30.8%–32.8%; Figure 10). Errors decreased across all 216 countries and regions. The disparity between regions narrowed from 0.200 to 0.163, while the gap between income groups shrank from 0.191 to 0.098. Nevertheless, low-income economies continued to exhibit the highest average errors, indicating that the improvement in overall performance coexists with an income-related error gradient.

E Detailed Geographic Analyses

E.1 Errors by Model and Indicator

Figure 11 reports the complete model–indicator error matrix evaluated over the recent reference window (2023–2025). Models and indicators are ordered by their overall mean ARE. The matrix reveals how aggregate performance decomposes across development domains: while modern models consistently achieve low error on macroeconomic aggregates such as population and GDP, substantial inaccuracies persist on human welfare metrics, particularly maternal mortality and education expenditure.

E.2 Regional and Income-Group Orderings

Figure 12 details the regional and income-group rankings corresponding to Section 3.5, where Rank 1 denotes the lowest error. Regional groupings follow World Bank classifications: NA (North America), ECA (Europe & Central Asia), LAC (Latin America & Caribbean), SA (South Asia), MENA (Middle East & North Africa), EAP (East Asia & Pacific), and SSA (Sub-Saharan Africa). Across both model generations and individual indicators, the rankings reveal a bifurcated hierarchy: Western dominance remains entrenched at the top (NA and ECA), while performance variance is heavily dispersed across peripheral regions, and low-income economies consistently occupy the lowest accuracy tier.

E.3 Countries with Persistently High Error

Figure 13 examines error persistence by tracking countries that consistently fall within the highest quartile of errors across model generations. Notably, 64.8% of these countries (35 out of 54) remained in the highest error tier throughout; furthermore, the ten countries exhibiting the largest errors in recent models were already present in the highest error quartile of earlier models. Crucially, seven of these ten most persistent high-error countries are Pacific Small Island Developing States (American Samoa, Tuvalu, Nauru, Northern Mariana Islands, Micronesia, Palau, and Marshall Islands), alongside the Faroe Islands, Djibouti, and Yemen. While absolute errors for these island economies decreased modestly (e.g., Tuvalu's ARE fell by 9.3%, and Nauru by 16.4%), their progress lagged far behind the global average. In contrast, several continental African nations achieved substantial gains and successfully exited the highest-error quartile, led by Uganda (55.1% reduction), Sudan (52.2%), and Malawi (50.8%). This difference suggests that while increasing the size of the model alleviates the fact gap problem in regions with moderate data representation, it marginalizes or even forgets economies with low levels of digitalization and in isolation at the algorithmic level.

E.4 Country Characteristics and Error

Figure 14 assesses how country-level error associates with national socioeconomic characteristics using Pearson correlation coefficients ($n = 123$–216, retaining original units). Across model generations, recall errors correlate positively with maternal mortality and negatively with electricity access, labor productivity (GDP per worker), and female parliamentary representation. Importantly, while the positive correlation with maternal mortality weakened from 0.597 to 0.452, negative correlations with total GDP (shifting from -0.218 to -0.287) and population (from -0.045 to -0.197) strengthened notably.

F Source Attribution and Numerical Reliability

Following Moayeri et al. (2024), we examine whether explicit citations to statistical authorities reflect genuine factual grounding or spurious attribution. Figure 15 presents three representative Jais-2-8B responses audited against annual statistics from 2016 to 2025. These examples reveal two primary modes of attribution failure. First, models can generate acute factual hallucinations despite citing authoritative sources. When asked about Papua New Guinea's unemployment rate, Jais-2-8B cited a 2023 World Bank estimate of 10.1%, whereas the true value was 2.7% (ARE $= 0.738$), and the error remained at 0.700 even when checked against the closest historical year in the decade-long verification window. Second, models frequently suffer from temporal misalignment by mislabeling outdated historical figures with recent years. For Montenegro's GDP, the model reported a 2022 figure of \$4.322 billion (ARE $= 0.310$ against 2022 ground truth), yet this exact figure matched Montenegro's 2016 GDP with an ARE of only 0.008. Similarly, for Madagascar's GDP, the model cited recent official estimates that closely matched 2016 data (ARE $= 0.024$) rather than contemporary values (ARE $= 0.316$). These cases demonstrate that explicitly citing sources can create an illusion of reliability, thereby masking a potential temporal hallucination, whereby the model retrieves outdated historical data while falsely presenting it as current information.

Table 1: **Indicator definitions and recent-reference coverage.** The final column counts countries and territories with at least one valid observation in the 2023–2025 window.

Indicator	Definition and unit	Countries
Agricultural land	Arable land, permanent crops, and permanent pasture as a share of land area (%).	208
CO_2 emissions per capita	CO_2 emissions excluding land use, land-use change, and forestry, divided by population (tonnes CO_2 equivalent per person; WDI EDGAR series).	202
Education expenditure	Total government expenditure on education as a share of GDP (%).	123
Access to electricity	Population with access to electricity as a share of total population (%).	214
GDP	Gross domestic product in current US dollars.	203
GDP per person employed	GDP per employed person in constant 2021 PPP international dollars.	171
Maternal mortality	Maternal deaths per 100,000 live births, using modeled estimates.	193
Population	Total midyear population, in persons.	216
Renewable energy	Renewable energy consumption as a share of total final energy consumption (%).	212
Unemployment	Unemployed persons as a share of the labor force, using modeled ILO estimates (%).	183
Women in parliament	Seats held by women in the single or lower chamber of the national parliament (%).	187

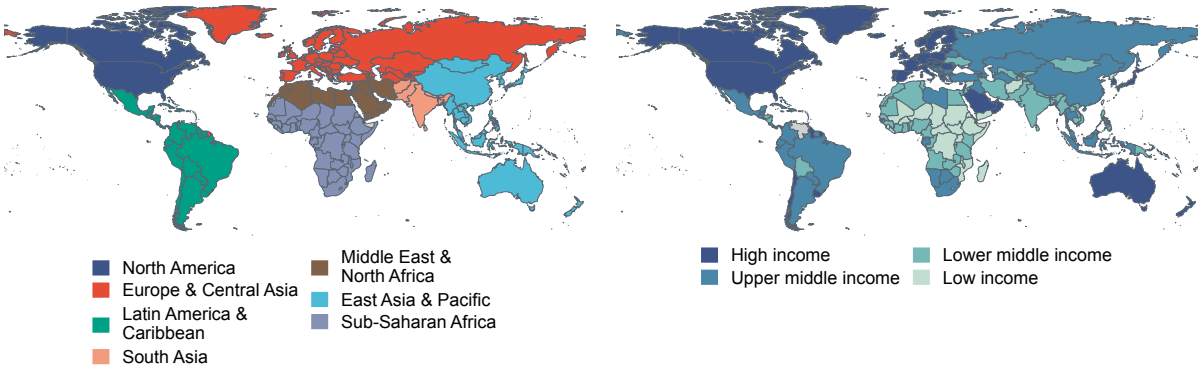


Figure 9: **Region and income-group classifications.** Gray denotes countries without the relevant assignment or outside the mapped inventory.

Table 2: **Model inventory: 14 earlier and 19 recent models.** Version gives the named checkpoint or registered service release; full revisions remain in the data inventory. Local inference uses vLLM or Transformers; hosted inference uses API.

Group	Model	Provider	Release	Version	Inference pipeline
Earlier models	**Open-weight models**				
	Qwen 14B Base	Alibaba	2023-09-25	Qwen-14B	Transformers
	Qwen 14B Chat	Alibaba	2023-09-25	Qwen-14B-Chat	Transformers
	Qwen 7B Base	Alibaba	2023-08-03	Qwen-7B	Transformers
	Qwen 7B Chat	Alibaba	2023-08-03	Qwen-7B-Chat	Transformers
	Zephyr 7B	Hugging Face H4	2023-10	zephyr-7b-beta	vLLM
	Vicuna 13B	LMSYS	2023-08	vicuna-13b-v1.5	vLLM
	Vicuna 7B	LMSYS	2023-08	vicuna-7b-v1.5	vLLM
	Llama 2 13B Chat	Meta	2023-07-18	Llama-2-13b-chat-hf	vLLM
	Llama 2 13B Base	Meta	2023-07-18	Llama-2-13b-hf	vLLM
	Llama 2 7B Chat	Meta	2023-07-18	Llama-2-7b-chat-hf	vLLM
	Llama 2 7B Base	Meta	2023-07-18	Llama-2-7b-hf	vLLM
	Orca-2-13B	Microsoft	2023-11-20	Orca-2-13b	vLLM
	Orca-2-7B	Microsoft	2023-11-20	Orca-2-7b	vLLM
	Mistral 7B	Mistral AI	2023-12-11	Mistral-7B-Instruct-v0.2	vLLM
Recent models	**Open-weight models**				
	Qwen2.5-7B	Alibaba	2024-09-19	Qwen2.5-7B-Instruct	vLLM
	Qwen3-8B	Alibaba	2025-04-29	Qwen3-8B	vLLM
	Qwen3.5-27B	Alibaba	2026-02-24	Qwen3.5-27B	vLLM
	Qwen3.5-9B	Alibaba	2026-03-02	Qwen3.5-9B	vLLM
	Command R	Cohere	2024-08-30	Command R (08-2024)	vLLM
	DeepSeek-Chat	DeepSeek	2024-05	DeepSeek-Chat	API
	Jais 2 8B	Inception / Cerebras / MBZUAI	2025-12-09	Jais-2-8B-Chat	vLLM
	Llama 3 8B	Meta	2024-04-18	Llama-3-8B-Instruct	vLLM
	Llama 3.1 8B	Meta	2024-07-23	Llama-3.1-8B-Instruct	vLLM
	Llama 4 Scout	Meta	2025-04-05	Llama-4-Scout-17B-16E-Instruct	API
	Phi-4	Microsoft	2024-12-12	phi-4	vLLM
	Falcon H1 7B	TII	2025-05-21	Falcon-H1-7B-Instruct	vLLM
	Proprietary models				
	Qwen3-Max	Alibaba	2025-09-24	qwen3-max	API
	Claude Opus 4.5	Anthropic	2025-11-24	Claude Opus 4.5	API
	Claude Sonnet 5	Anthropic	2026-06-30	claude-sonnet-5-20260630	API
	Gemini 2.5 Pro	Google	2025-06-17	gemini-2.5-pro	API
	GPT-5.2	OpenAI	2025-12-11	gpt-5.2	API
	Solar Pro 4	Upstage	2026-08	solar-pro4	API
	Grok-4	xAI	2025-07-09	grok-4	API

Table 3: **Per-model error and evaluable coverage.** Coverage denominators are 2,112 questions and 2,131 specific-year questions. Dashes denote conditions absent from the analysis. The final column marks completed web-search results.

Model	Mean ARE	Coverage	2021 ARE	Coverage	Web
Earlier models					
Qwen 14B Base	0.478	98.1%	0.453	99.1%	No
Qwen 14B Chat	0.471	100.0%	0.399	99.8%	No
Qwen 7B Base	0.497	98.1%	0.522	98.8%	No
Qwen 7B Chat	0.507	100.0%	0.465	90.3%	No
Zephyr 7B	0.567	100.0%	0.371	93.8%	No
Vicuna 13B	0.414	100.0%	0.371	98.9%	No
Vicuna 7B	0.474	100.0%	0.463	99.1%	No
Llama 2 13B Chat	0.492	100.0%	0.461	100.0%	No
Llama 2 13B Base	0.544	99.9%	–	–	No
Llama 2 7B Chat	0.545	100.0%	0.438	97.7%	No
Llama 2 7B Base	0.558	95.9%	–	–	No
Orca-2-13B	0.494	99.2%	0.416	99.8%	No
Orca-2-7B	0.494	99.9%	0.496	98.3%	No
Mistral 7B	0.472	99.1%	0.384	79.5%	No
Recent models					
Qwen2.5-7B	0.380	100.0%	0.361	100.0%	No
Qwen3-8B	0.414	100.0%	0.426	99.9%	No
Qwen3.5-27B	0.307	99.2%	0.263	100.0%	No
Qwen3.5-9B	0.423	100.0%	0.400	100.0%	Yes
Command R	0.295	100.0%	0.274	99.8%	No
DeepSeek-Chat	0.167	99.9%	0.149	99.6%	Yes
Jais 2 8B	0.464	78.8%	0.519	97.4%	No
Llama 3 8B	0.418	100.0%	0.380	100.0%	No
Llama 3.1 8B	0.357	100.0%	0.361	99.6%	No
Llama 4 Scout	0.282	100.0%	0.255	99.9%	Yes
Phi-4	0.298	87.1%	0.277	96.8%	No
Falcon H1 7B	0.389	99.6%	0.362	99.9%	No
Qwen3-Max	0.295	100.0%	–	–	Yes
Claude Opus 4.5	0.172	99.5%	–	–	No
Claude Sonnet 5	0.182	100.0%	0.145	99.4%	No
Gemini 2.5 Pro	0.172	99.5%	0.143	99.6%	Yes
GPT-5.2	0.178	99.8%	0.165	100.0%	Yes
Solar Pro 4	0.273	100.0%	–	–	No
Grok-4	0.180	99.7%	–	–	No

Table 4: **Agricultural-land prompts for the main and specific-year experiments.** Switzerland provides the example; Aruba is the target country.

question

```
SYSTEM:
I will ask you factual questions about
countries. Specifically, I will ask you for
the percent of total land area that is
agricultural. You will answer as concisely as
possible - only answer with the number! First
I will give an example with the answer. Then
I will ask you my question, and you will
provide the answer in the same way.

USER:
What is the percent of total land area that
is agricultural for the country Switzerland?
Do not answer in a complete sentence - only
provide the number!

ASSISTANT:
37.988

USER:
What is the percent of total land area that
is agricultural for the country Aruba?  Do
not answer in a complete sentence - only
provide the number!
```

Question specifying 2021

```
USER:
What was the agricultural land of Switzerland
in 2021? Give the value in percent of land
area (0-100). Report only the numerical value,
without an explanation.

ASSISTANT:
37.9496922648985

USER:
What was the agricultural land of Aruba in
2021? Give the value in percent of land area
(0-100). Report only the numerical value,
without an explanation.
```

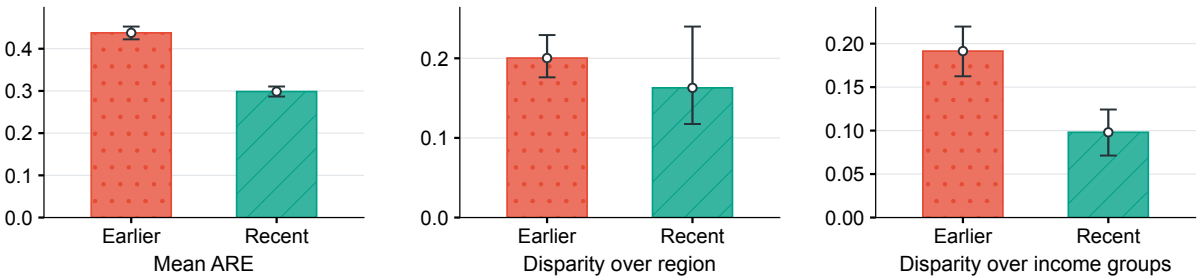


Figure 10: **Mean ARE and group disparities for the year-specific (2021) task, comparing 12 earlier and 15 recent models.** Whiskers show 95% country bootstrap intervals.

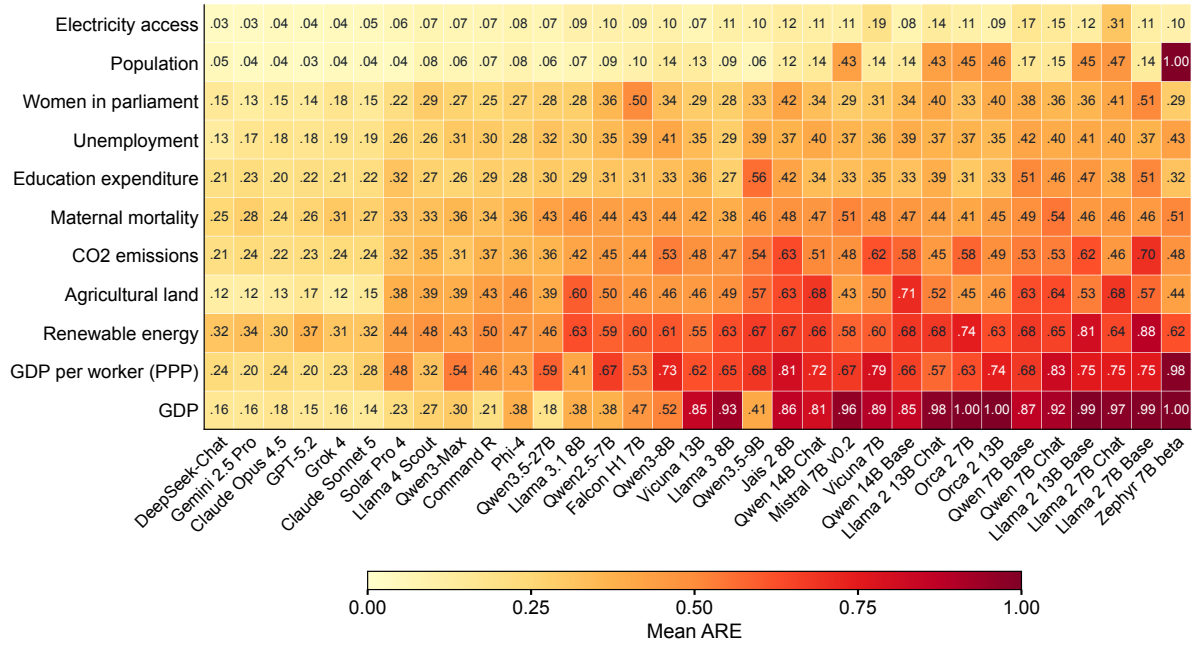


Figure 11: **Mean ARE by model and indicator, sorted by overall error.** Cells show two-decimal values; darker colors indicate higher error.

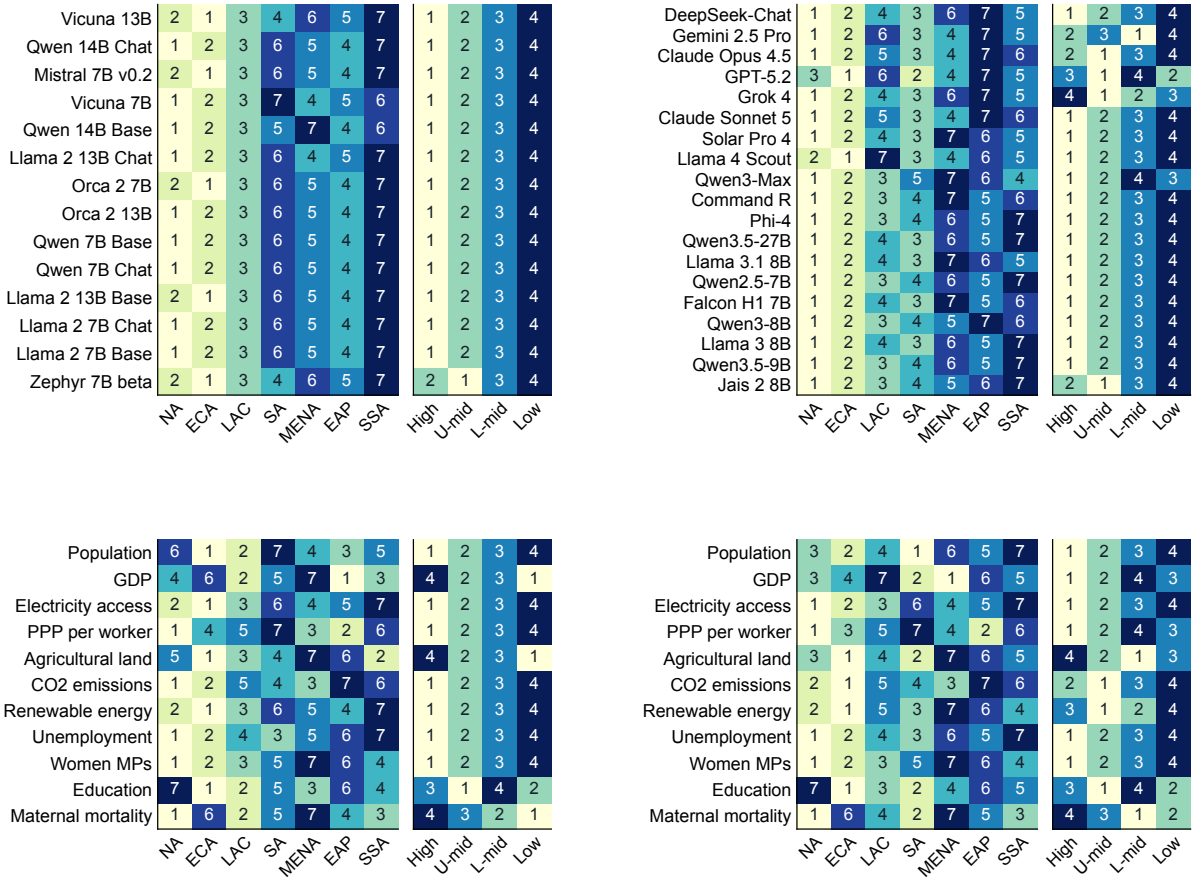


Figure 12: **Group error ranks within models (top) and indicators (bottom), for earlier (left) and recent (right) models.** Rank 1 indicates the lowest error. Abbreviations are defined in Appendix E.2.

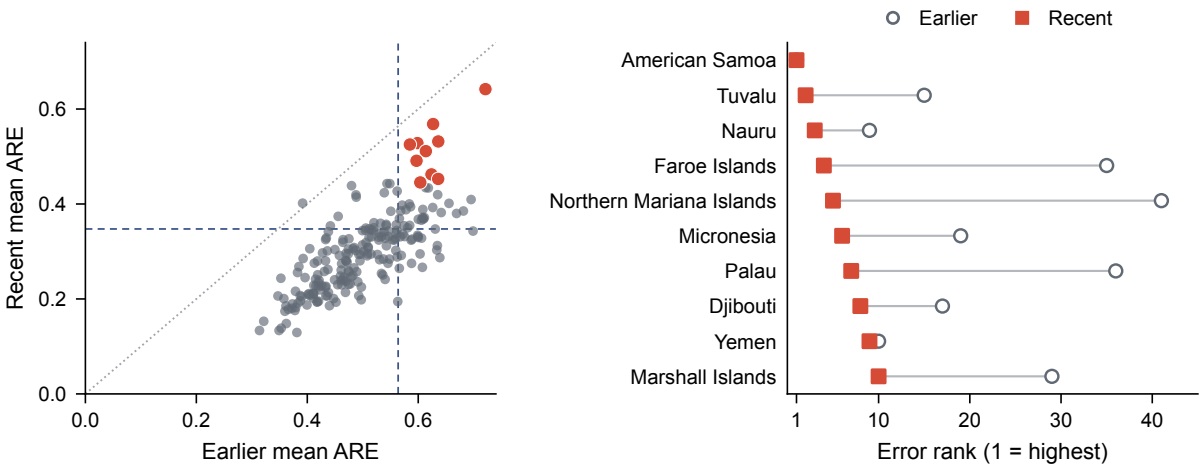


Figure 13: **Country errors and persistence in the highest-error quartile.** Red points mark the ten highest recent errors; the right panel tracks their error ranks. Rank 1 indicates the highest error.

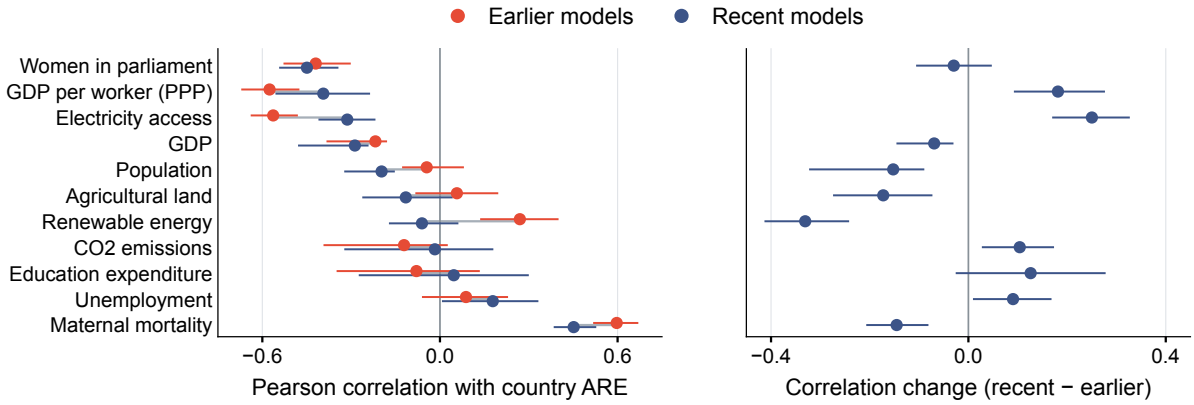


Figure 14: **Pearson correlations of country characteristics with mean ARE and their changes.** Whiskers show 95% paired country bootstrap intervals; each characteristic uses its available countries.

Papua New Guinea

Unemployment · Jais 2 8B

Unemployment rate in Papua New Guinea is estimated to be around 10.1%.

Source: World Bank, 2023 estimate.

Recent-window ARE: 0.741
Cited-year ARE: 0.738
Best 2016–2025 ARE: 0.700
Closest year: 2021

Montenegro

GDP · Jais 2 8B

Montenegro's nominal GDP in 2022 was estimated at 4.322 billion USD.

(Source: World Bank, October 2023 Estimate - Subject to Change)

Recent-window ARE: 0.484
Cited-year ARE: 0.310
Best 2016–2025 ARE: 0.008
Closest year: 2016

Madagascar

GDP · Jais 2 8B

Madagascar's GDP (current US dollars) is estimated to be around 12.14.

[…]

World Bank and the International Monetary Fund (IMF) […] USD 12-14 billion for recent years

Recent-window ARE: 0.316
Cited year: not specified
Best 2016–2025 ARE: 0.024
Closest year: 2016

Figure 15: **Three Jais-2-8B source-attribution examples, checked against recent, claimed-year, and nearest-year reference values.** The cases illustrate numerical and temporal mismatches.